DNA Sequence Analysis Report

Prepared 2026-11-08T07:31:21.945Z

Homo sapiens HBB coding fragment, NM_000518.5 coding nt 1-117

Length: 117 nt · **Genetic code:** table 1 (standard)

ORF rules: ATG start; stops TAA, TAG, TGA; minimum 30 nt including the stop codon. Complete candidates only. Coordinates are 1-based inclusive on the original input.

Sequences are processed on this device. Candidate ORFs and translated frames are computational results, not evidence of gene function or expression.

Nucleotide composition

Base or category	Count (nt)	Percentage of A/C/G/T
A	19	16.24%
C	26	22.22%
G	44	37.61%
T	28	23.93%
Ambiguous IUPAC symbols	0	Excluded from known-base percentages

GC: 59.83% · **AT:** 40.17%. Denominator: 117 unambiguous A/C/G/T bases; 0 ambiguous symbols remain in the full input.

Short-oligo Wallace-rule Tm estimate: Unavailable. Only reported for unambiguous 4–14 nt oligonucleotides. This approximate rule omits salt, concentration and nearest-neighbour effects.

Input DNA · 5′ → 3′

```
ATGGTGCATCTGACTCCTGAGGAGAAGTCTGCCGTTACTGCCCTGTGGGGCAAGGTGAACGTGGATGAAGTTGGTGGTGAGGCCCTGGGCAGGC
TGCTGGTGGTCTACCCTTGGACC
```

Strand transformations

Complement aligned to input · 3′ → 5′

```
TACCACGTAGACTGAGGACTCCTCTTCAGACGGCAATGACGGGACACCCCGTTCCACTTGCACCTACTTCAACCACCACTCCGGGACCCGTCCG
ACGACCACCAGATGGGAACCTGG
```

Reverse complement · 5′ → 3′

```
GGTCCAAGGGTAGACCACCAGCAGCCTGCCCAGGGCCTCACCACCAACTTCATCCACGTTCACCTTGCCCCACAGGGCAGTAACGGCAGACTTC
TCCTCAGGAGTCAGATGCACCAT
```

Coding-strand RNA substitution · 5′ → 3′

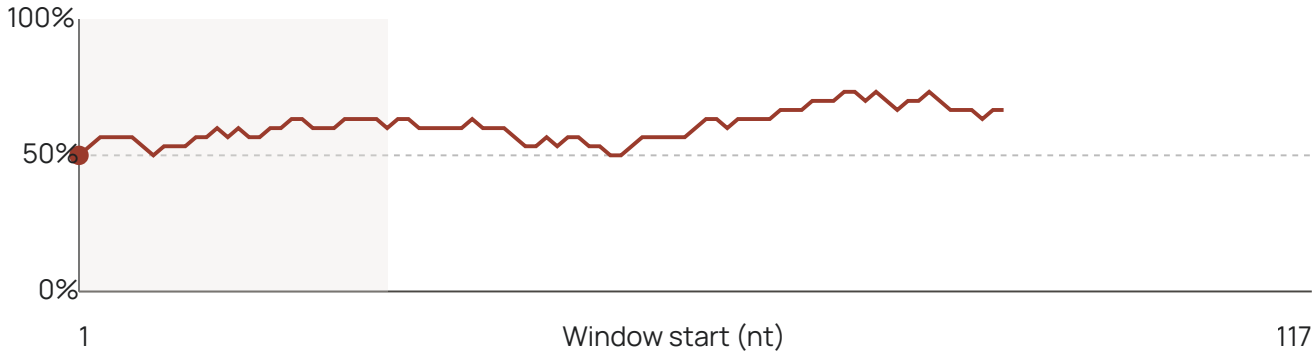
```
AUGGUGCAUCUGACUCCUGAGGAGAAGUCUGCCGUUACUGCCCUGUGGGGCAAGGUGAACGUGGAUGAAGUUGGUGGUGAGGCCCUGGGCAGGC
UGCUGGUGGUCUACCCUUGGACC
```

RNA from input treated as a 5′ → 3′ template strand

```
GGUCCAAGGGUAGACCACCAGCAGCCUGCCCAGGGCCUCACCACCAACUUCAUCCACGUUCACCUUGCCCCACAGGGCAGUAACGGCAGACUUC
UCCUCAGGAGUCAGAUGCACCAU
```

Sliding-window GC profile

Window: 30 nt. Positions on the horizontal axis are window starts in the input.



Each point is one complete window, advanced by one nucleotide. Ambiguous symbols are excluded from each window's denominator. GC composition alone does not identify biological function.

Six-frame translation

Each oriented DNA strand is read 5′ → 3′. Reverse frames use the reverse complement; offsets count from that strand's 5′ end. * denotes a stop signal; X denotes an unresolved amino acid. Incomplete terminal codons are omitted.

+1 · forward · offset 0 nt · 0 terminal nt omitted

```
MVHLTPEEKSAVTALWGKVNVDEVGGEALGRLLVVYPWT
```

+2 · forward · offset 1 nt · 2 terminal nt omitted

```
WCI*LLRRSLPLLPCGAR*TWMKLVVRPWAGCWWSTLG
```

+3 · forward · offset 2 nt · 1 terminal nt omitted

```
GASDS*GEVCRYCPVGQGERG*SWW*GPGQAAGGLPLD
```

−1 · reverse complement · offset 0 nt · 0 terminal nt omitted

```
GPRVDHQQPAQGLTTNFIHVHLAPQGSNGRLLLRSQMHH
```

−2 · reverse complement · offset 1 nt · 2 terminal nt omitted

```
VQG*TTSSLPRASPPTSSTFTLPHRAVTADFSSGVRCT
```

−3 · reverse complement · offset 2 nt · 1 terminal nt omitted

```
SKGRPPAACPGPHHQLHPRSPCPTGQ*RQTSPQESDAP
```

Candidate open reading frames

0 complete candidates at the selected threshold. Lengths include the terminal stop codon; peptides retain its * symbol.

No complete candidates meet these rules. This does not establish that the sequence is noncoding.

Codon usage · +1

39 unambiguous codons counted; 0 ambiguous codons excluded from usage frequencies.

DNA codon	Amino acid	Count	Frequency
AAC	N	1	2.56%
AAG	K	2	5.13%
ACC	T	1	2.56%
ACT	T	2	5.13%
AGG	R	1	2.56%
ATG	M	1	2.56%
CAT	H	1	2.56%
CCT	P	2	5.13%
CTG	L	5	12.82%
GAA	E	1	2.56%
GAG	E	3	7.69%

DNA codon	Amino acid	Count	Frequency
GAT	D	1	2.56%
GCC	A	3	7.69%
GGC	G	2	5.13%
GGT	G	2	5.13%
GTC	V	1	2.56%
GTG	V	4	10.26%
GTT	V	2	5.13%
TAC	Y	1	2.56%
TCT	S	1	2.56%
TGG	W	2	5.13%

Motifs and repeated patterns

Search the current sequence to locate a motif.

Reverse-complement palindromes

Exact A/C/G/T matches, 4–12 nt.

`5–8 · TGCA` `59–62 · ACGT` `81–84 · GGCC`

Tandem repeats

Exact adjacent copies, unit size 1–6 nt, at least 3 copies. Up to 12 example intervals per pattern class are shown.

`41–43 · C × 3` `47–50 · G × 4` `83–85 · C × 3` `87–89 · G × 3` `108–110 · C × 3`